ENGINEERING PROJECT MANAGEMENT

EMGT 5220 – Fall 2022

Project Proposal Draft 1

Team 1

Authors:

Gabriele Ansaldo
Ole Martin Kjorholt
Jignasuben Rohitbhai Vekariya
Prachi Patel
Jayanth Reddy Rachamallu
Kshitija Sonavane

Team's Point of Contact:

Gabriele Ansaldo
ansaldo.g@northeastern.edu

October 19, 2022

Table of Contents

Letter of Transmittal **i**

Executive Summary **ii**

1 Introduction **1**

1.1 Problem 1

1.2 Solution 1

2 Purpose & Objectives **2**

2.1 Purpose 2

2.2 Objectives 2

3 Technical Overview **3**

4 Implementation Plan **4**

4.1 WBS 4

4.2 Schedule 4

4.3 Responsibility Chart 4

4.4 Resource Allocation 4

4.5 Stakeholders 4

5 Execution Plan **5**

5.1 Project Monitoring 5

5.2 Project Control 5

5.3 Project Auditing 5

5.4 Project Closure 5

6 Risk Assessment Management Plan **6**

7 Financial Plan with Budget **7**

7.1 High Level Details 7

7.2 Budget Justification 7

8 Team Credentials **8**

Bibliography **10**

Appendices **11**

Letter of Transmittal

Executive Summary

1 Introduction

1.1 Problem

The city of Boston has long been plagued by traffic problems. Throughout the years, various projects such as the Big Dig have been carried out to ease road congestion and improve traffic flow (Serna, 2019). Although traffic levels remain high in the city and surrounding region, it is evident that roadway construction is limited. Nevertheless, new roads are costly, stimulate induced demand, and negatively affect the environment. Even with recent and substantial investments, Boston ranked consistently within the top ten congested cities in the United States (Friedman, 2020). Congestion has major impacts on economic growth and productivity, and it leads to increased fuel consumption, emission, and reduced road safety for both drivers and pedestrians. Due to high traffic levels, a recent phenomenon that began to emerge is the one of cut-through driving, which is a tactic used to avoid heavy traffic and long delays, whereas drivers take secondary routes. Once again, this phenomenon aggravates the already existing congestion and it contributes to other traffic jam on the cut-through streets, along with accompanying problems such as collisions, pollution from exhaust, or road rage. According to projections, the population of Boston will grow by approximately 15 - 17% by 2030, reaching 710,000 to 724,000 people (City of Boston, 2017). This indicates that high demands will be placed on the road system, and an early solution is necessary.

1.2 Solution

When it comes to traffic control, the use of traffic lights is one of the major components. Several studies have demonstrated that traffic lights can be used efficiently to reduce congestion and all its associated issues. There is no doubt that traffic is a major issue affecting the city of Boston. Therefore, for the purpose of alleviating traffic congestion, this project proposes the implementation of novel sensor technologies and Artificial Intelligence (AI) optimization tools. Specifically, it is proposed to implement a Smart Adaptive Traffic Control System (SATCS) to efficiently control traffic lights. This type of system would allow the collection of traffic data around traffic lights, such as the type, number, and waiting time of vehicles. Through the use of this data, Machine Learning (ML) techniques can be used to optimize the control of traffic lights to reduce traffic congestion throughout the city. Upon completion of the project, the team will have installed SATCS devices throughout Boston. In particular, it is intended to create the next generation of integrated traffic signal systems that include rapid simulation of future traffic conditions. This system is also expected to reduce crashes and pollution as well as balance road-user delays. To realize such a project successfully, an interdisciplinary team of traffic, software, and civil engineers will work in concert with the project manager. Most of the engineers working on this project are from the city of Boston, which means that they are familiar with the city and the issues associated with traffic. As the Project Manager hired by the City of Boston has previously managed similar projects, he or she will be able to guide the project smoothly and will be aware of any issues that might arise. This project has great potential for both the city and its residents. As a result of implementing this system, drivers' fuel consumption, state fuel consumption, delivery times, shipping costs, mechanical failures of cars caused by stop-and-go traffic, and employee turnover could all be reduced.

2 Purpose & Objectives

2.1 Purpose

The purpose of this project is to improve the road infrastructure of the city of Boston by reducing traffic congestion with the use of a SATCS, which ultimately will maximize the use of existing roadway capacities while allowing for smoother traffic, reduction in fuel consumption and noise and air pollution, and increased road safety for both drivers and pedestrians. In particular, this project aims to develop and install a SATCS on 70% of the traffic lights in the Boston area within eight months. With the aid of AI and ML, this system will continuously monitor traffic and control traffic lights in real time to optimize traffic flow. With the integration of this technology, Boston will be able to meet the increased demand to the road system that is expected to occur as a result of a substantial population increase over the next few years.

2.2 Objectives

The objective of this project is to deploy the SATCS system on 70% of the 840 traffic lights within the city of Boston. Through the use of advanced traffic control strategies, the SATCS project will enable dynamic traffic routing to maximize the utilization of existing roadway capacity. As part of the SATCS project, a suite of modeling tools and methods will be required to implement AI routing techniques. A dynamic and proactive evaluation of AI strategies using real-time and historical data will be possible through the application of these techniques, tools, and methods by the city of Boston. Following is a list of objectives that are based on the purpose and goal of this project:

Goals	Objectives
Reduce Traffic Congestion	- Reduce waiting at intersections and traffic lights
	- Higher traffic management ability
Reduce Environmental Impact	- Reduce noise pollution
	- Reduce air pollution and vehicle emissions
Improve Mobility	- Reduce travel time for drivers
	- Reduce fuel consumption and costs
	- Reduce travel stress
Increase Safety	- Faster detection of accidents
	- Pedestrians and drivers' safety monitored

Table 1: Goals and Respective Objectives of SATCS

3 Technical Overview

4 Implementation Plan

4.1 WBS

4.2 Schedule

4.3 Responsibility Chart

4.4 Resource Allocation

4.5 Stakeholders

5 Execution Plan

5.1 Project Monitoring

5.2 Project Control

5.3 Project Auditing

5.4 Project Closure

6 Risk Assessment Management Plan

7 Financial Plan with Budget

7.1 High Level Details

7.2 Budget Justification

8 Team Credentials

Gabriele Ansaldo

Gabriele is a graduate student at Northeastern University studying Engineering Management. Previously, he earned a Master's degree in Mechanical Engineering from TU Delft and a Bachelor's degree in Industrial Engineering & Management from the University of Groningen. He spent a year as a Research Student at Northeastern University at the MAGICS Lab focusing on reinforcement learning and robotics. His previous experience includes co-founding a start-up focused on developing IoT devices to help people monitor the effects of their actions on the environment. Through his startup experience, he was able to translate engineering skills into business and industry language. Particularly, he developed qualities such as discipline, time management, leadership, and commitment.

Ole Kjørholt

Ole is a graduate student pursing Engineering Management at Northeastern University. Prior to this, he graduated from the University of Central Arkansas with a Bachelor of Science in Engineering Physics. He has experience in robotics and has completed internships related to software development and automation. Besides his technical expertise, he also served as captain of both the Division 1 soccer teams at his universities, which has provided him with invaluable experience in leading and working as a member of a team.

Jignasuben Rohitbhai Vekariya

Jignasu is a graduate student at Northeastern University seeking a master's degree in data analytics engineering. She graduated from Veer Narmad South Gujarat University with a Bachelor of Science in Science, with a major in Computer Science, prior to this. Throughout her undergraduate education, she developed software, took part in workshops, and ran competitions to broaden her technological skills. Additionally, she enjoys volunteering, which boosts her confidence. She organized a seven-day social work camp for college students in addition to extracurricular activities like competitions and workshops, and she volunteered for three days at a camp to advertise her college. She presently contributes to Origin, a student group for data analytics engineers, and she goes to workshops focused on research to advance her career. Through her voyage, she gained skills in leadership, teamwork, discipline, and strength.

Prachi Patel

Prachi is a graduate student at Northeastern University pursuing a master's degree in Data Analytics Engineering. Previously, she worked as a software engineer at an aerospace firm after interning there for six months where her major project involved optimizing video surveillance systems for commercial aircraft. Among her other endeavors are internships with an educational services firm and technical training at HAL, both of which she completed during her studies at MS Ramaiah Institute of Technology for her Bachelor of Science in Electronics and Instrumentation Engineering. Besides acquiring core technical skills, her experience as an intern and as a full-time employee has taught her valuable lessons about time management, team spirit, perseverance, and diligence.

Jayanth Reddy Rachamallu

Jayanth is a graduate student at Northeastern University pursuing his master's in Data Analytics Engineering. Prior to this, he earned a Bachelor of Technology from the Sreenidhi Institute of Science and Technology. With a background in Computer Science and Engineering, he has been an intern for six months developing a mobile application. In the final semester of his undergraduate studies, he worked on a major project related to IOT. In addition to enhancing his technical skills, these experiences have taught him about time management, team spirit, and commitment.

Kshitija Sonavane

Kshitija is a graduate student at Northeastern University pursuing her master's in Industrial Engineering. She holds a Bachelor of Technology degree in mechanical engineering from Vellore Institute of Technology. During her undergraduate studies, she worked as a project manager for SEDS INDIA, a student chapter organization established by MIT USA. The process of transforming an idea into reality has always been of great interest to her. In addition, she was a member of her university's robotics team and participated in competitions such as the European Rover Challenge (2nd position in Asia) and the University Rover Challenge, USA (19th place worldwide). Her commitment to giving back to society led her to participate in various outreach programs that taught science and astronomy to schoolchildren. Through these enriching experiences, she has been able to build a strong foundation for good technical and leadership skills.

Bibliography

City of Boston. (2017, March). *Boston In 2023.* Retrieved October 2022, from boston.gov: https://www.boston.gov/sites/default/files/file/document_files/2017/03/go_boston_2030_-_6_boston_in_2030_spreads.pdf

Friedman, J. (2020, October 13). *The 10 Most Congested Cities in the U.S.* Retrieved October 2022, from usnews.com: https://www.usnews.com/news/cities/articles/10-cities-with-the-worst-traffic-in-the-us

Serna, N. V. (2019). *Managing Traffic in Massachusetts: Assessing the Potential Income Equity Impacts of Congestion Pricing in Greater Boston.* Harvard University. Boston: Harvard Kennedy School.

Appendices